A2: SPC Control Chart Workbook

Context and Rationale

Manufacturing Setting: The Ubuntu Naturals Johannesburg tableting line (South Africa, Tier 1; Stage 4) runs **Hibiscus Pressure** tablets around the clock. The batch record SOP requires an in-process sample every two hours, measured for hardness and weight. Those measurements have been drifting slowly for weeks. Nobody has called it a problem yet, which is the position you are being asked to work from: complaint and out-of-spec rates are climbing, and a SAHPRA inspector has not yet had cause to write an observation.

Rationale: Variation in tablet hardness and weight comes back as customer complaints, returns from pharmacy retail, and audit findings against the batch record at SAHPRA or PIC/S inspection. Without control charts the line can only react: a problem announces itself through a return or a failed in-process check, long after the batch that caused it has shipped. A worked SPC workbook lets the Johannesburg team see drift while it is still drift.

Brief

You are given 120 tablet measurements from the Hibiscus Pressure tableting line, in 24 subgroups of five. Chart them, judge whether the process is stable, work out its capability and performance against the tablet specification limits, and write the out-of-control action plan the production team would actually pin up at the workstation.

Key Requirements

- Select and justify the appropriate control chart type for the given data (X-bar and R chart for subgrouped variable data)
- Construct the control charts with correctly calculated centre lines, UCL, and LCL using formulae (not just software output)
- Identify any out-of-control points using at least 4 of the standard out-of-control rules and explain what each pattern suggests about the process
- Calculate Cp, Cpk, Pp, and Ppk indices given the specification limits; interpret the difference between capability and performance
- Include a commentary worksheet explaining what the charts reveal about the process and recommending corrective actions
- Design a one-page out-of-control action plan (OCAP) suitable for posting at the operator workstation
- Assess the trade-off between 2-sigma warning limits and 3-sigma action limits for this process, considering the cost of false alarms vs. missed defects

- Design a 2-factor factorial experiment (e.g., testing the effect of compression force and dwell time on Hibiscus Pressure tablet hardness): define factor levels, construct the design matrix, specify the response variable and replication, and include ANOVA analysis with interpretation of main effects and interactions

Production Guidelines

1. **Examine the dataset**: Open `A02_hibiscus_pressure_spc_120.csv`, attached to this assignment -- 120 tablet measurements in 24 subgroups of 5, with the hardness and weight specification limits stated in its header. Check for missing values and transcription errors, and confirm the subgroup structure before you chart anything.
2. **Select and justify chart type**: Document why X-bar and R charts are appropriate for subgrouped variable data with n=5. Explain in a commentary cell why this choice is preferable to, say, Individuals/MR or X-bar/S charts for this subgroup size.
3. **Calculate control limits using formulae**: In a dedicated worksheet, show the calculation of the grand mean (X-double-bar), average range (R-bar), and control limits using the A2, D3, and D4 constants. Do not rely solely on software-generated charts -- the formulae must be visible and auditable.
4. **Construct the control charts**: Build X-bar and R charts with correctly plotted centre lines, UCL, and LCL. Use conditional formatting or markers to highlight out-of-control points.
5. **Apply out-of-control rules**: Test the data against at least 4 of the 8 Western Electric / Nelson rules (e.g., 1 point beyond 3-sigma, 2 of 3 beyond 2-sigma, 4 of 5 beyond 1-sigma, 8 consecutive points on one side). Annotate each detected signal with which rule triggered it and what process behaviour it suggests.
6. **Calculate capability indices**: Compute Cp, Cpk, Pp, and Ppk using the provided specification limits. Explain the difference between capability (within-subgroup variation) and performance (total variation) in the context of tablet hardness/weight consistency.
7. **Create the OCAP**: Design a one-page Out-of-Control Action Plan formatted for posting at the operator workstation. Include: trigger conditions, immediate actions, escalation contacts, and documentation requirements.
8. **Design a factorial experiment**: In a separate worksheet, design a 2-factor factorial experiment relevant to Hibiscus Pressure tablet quality. Define two controllable factors (e.g., compression force and dwell time), specify at least two levels for each, construct the design matrix with replication, and perform ANOVA analysis. Interpret main effects and interactions, and connect the findings to process improvement recommendations.
9. **Write the commentary worksheet**: Summarise what the charts reveal about the process (stable/unstable, capable/incapable), recommend corrective actions, integrate DoE findings into the improvement plan, and assess the 2-sigma vs. 3-sigma limit trade-off for this process.

Marking Scheme

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
Chart Construction and Formulae	20%	Charts missing or incorrectly constructed; control limits wrong; no visible formulae.	Charts present with correct centre lines and control limits; formulae shown but may contain minor errors; basic formatting.	Charts well-constructed with all formulae correct, visible, and clearly labelled; conditional formatting highlights out-of-control points; professional chart layout.	Charts are publication-quality; formulae are organised in a separate calculation worksheet with clear cell references; chart formatting aids rapid visual interpretation (colour-coded zones).	Workbook includes dynamic charts that update when new data is appended; includes helper formulae for all 8 Nelson rules with automated flagging; provides a template worksheet for ongoing production use.
Out-of-Control Analysis	20%	No attempt to apply out-of-control rules, or rules applied incorrectly.	At least 2 rules applied correctly; out-of-control points identified but interpretation is generic or missing.	At least 4 rules applied correctly; each detected signal is annotated with the specific rule and a plausible process explanation (e.g., "8 consecutive points above centre line suggests a sustained shift -- check tablet press punch wear").	All applicable rules tested systematically; interpretations are specific to tablet manufacturing (linking patterns to known causes: granulation batch changes, compression force drift, punch / die wear).	Includes a run-by-run log linking each out-of-control signal to a root cause hypothesis with supporting data; proposes a stratification analysis (by shift, granulation lot, or tablet press) to isolate assignable causes.
Capability Indices	20%	Cp/Cpk not calculated, or calculations contain fundamental errors; no interpretation.	Cp and Cpk correctly calculated; basic interpretation (capable/not capable) provided.	All four indices (Cp, Cpk, Pp, Ppk) correctly calculated; clear explanation of the distinction between capability and performance; results interpreted in context of tablet specifications.	Interpretation includes confidence intervals or sample-size considerations for the indices; discusses what the gap between Cp and Cpk (or Pp and Ppk) reveals about process centring; recommends a target Cpk for the tablet line.	Includes a process capability improvement simulation showing the impact of reducing variation or re-centring the process; benchmarks results against industry standards (e.g., Six Sigma targets); projects defect rates (PPM) from the calculated indices.

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
OCAP Design	15%	No OCAP, or a generic template with no adaptation to the tableting process.	OCAP present with basic structure (trigger, action, escalation) but generic; would not be immediately usable by an operator.	OCAP is clearly structured, visually formatted for workstation posting, and includes specific trigger conditions linked to the control chart rules; actions are concrete (e.g., "check upstream granulation batch changeover log").	OCAP is operator-ready: uses plain language, includes a visual flowchart or decision tree, specifies escalation by role (operator, shift supervisor, quality engineer), and includes documentation requirements.	OCAP includes response time targets, links each rule violation to a most-likely-cause checklist specific to tablet compression, and proposes a periodic review cycle for updating the plan based on accumulated findings.
Design of Experiments (DoE)	10%	No DoE component, or experiment design is fundamentally flawed (e.g., factors and responses confused, no replication).	A 2-factor experiment is defined with factor levels and response variable identified; basic design matrix present but ANOVA analysis is missing or incorrect.	2-factor factorial design correctly specified with appropriate factor levels for tablet manufacturing (e.g., compression force and dwell time); design matrix complete with replication; ANOVA analysis correctly performed with main effects and interaction interpreted.	ANOVA results interpreted in the tablet manufacturing context: explains what a significant interaction means for production practice; connects DoE findings to the SPC charts (e.g., "if dwell time drives hardness, monitor dwell time with a separate control chart"); includes effect plots (main effects and interaction).	Extends DoE to include a response surface or optimisation step; proposes a sequential experimentation strategy (screening design followed by optimisation); discusses practical constraints of running experiments on a 24/7 tableting line (blocking, split-plot considerations); connects DoE results to process capability improvement targets.
Commentary and Risk Assessment	15%	No commentary, or commentary is a restating of numbers without interpretation or recommendation.	Commentary summarises chart findings and states whether the process is in control and capable; basic recommendations provided.	Commentary provides a coherent narrative linking chart patterns to process behaviour; 2-sigma vs. 3-sigma trade-off discussed with reference to false alarm and missed defect rates.	Commentary quantifies the cost implications of the current process performance (e.g., estimated reject rate, customer complaint risk); 2-sigma vs. 3-sigma analysis includes a cost-benefit calculation for the Ubuntu tablet context.	Commentary proposes a phased SPC implementation roadmap for the plant (pilot line to full Johannesburg deployment); discusses how SPC integrates with the plant's broader quality management system; addresses cultural and training barriers to SPC adoption on the production floor.

Stretch Goal

An exceptional submission would extend the workbook into a reusable SPC monitoring template that the Ubuntu Johannesburg tableting team could deploy immediately: dynamic charts that auto-update as new subgroups are entered, automated rule-checking with colour-coded alerts, a trends dashboard summarising weekly/monthly capability evolution, and a training sheet explaining how operators should use the workbook during each shift. The commentary would include a cost-of-poor-quality calculation estimating the financial impact of the current variation level and the projected savings from achieving the recommended capability target.

Submitting This Artefact

What you hand in: Excel/LibreOffice Calc workbook with multiple worksheets, embedded control charts, capability calculations, and interpretation commentary.

This is a Phase 1 (Foundational) artefact. It builds on programme entry, and the work you do here carries into A3 (SPC knowledge applied programmatically in Python) -- so it is worth doing properly the first time rather than twice.

Two files are attached to the assignment: this brief, as `A02_spc_control_chart_workbook.pdf`, and the dataset it refers to, as `A02_hibiscus_pressure_spc_120.csv`. A worked exemplar from another Ubuntu Naturals site is attached too, with the marker's note on it, and `A02_genai_guide.pdf` shows how to build your own submission with a free Claude account.

Name the file `A02_SPC_Workbook_[YourSurname].xlsx`, with your own surname in place of `[YourSurname]` -- the portfolio guide uses this convention throughout, and a marker looking for your work will look for that name.

Whatever else it contains, the file needs labelled worksheets, visible formulae (not hidden), a "README" sheet explaining the workbook.